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Smart Signal – Adaptive Traffic Signal Control using Reinforcement Learning and Object Detection

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Abstract. In order to solve the problems that reduce traffic efficiency in the road network, which is caused by overlarge traffic demand of urban regions at peak time, where traditional traffic light systems are unable to handle, we are proposing a working model of smart adaptive traffic signal control system based on single-agent reinforcement learning algorithm. Our system uses object detection algorithm to sense real-time traffic scenario, which is used by the reinforcement learning algorithm to compute optimal signal timing for the given scenario. Simulation results show that our model has good practicability and is substantially cheaper than the systems currently in use.

Keywords: Reinforcement Learning, Object Detection, Traffic Signal, Adaptive Traffic Signal Control, Cloud Computing, Microcontroller

1. Introduction

Present day population growth is expeditious. In a flourishing economy there is extensive need of transportability. The advancement in the social and economic activities outperform the transportation infrastructure which leads to clotting at intersections and connected roads. As a result many regions face the problem of traffic congestion and the routine tasks of the economy are affected. Almost every major city in India face this problem. Kolkata, Delhi, Mumbai and Bangalore are some of the major cities facing congestion and costs the government more than a lakh rupees annually. This also deteriorates the environment and decreases the standard of living. Ordinary traffic signals cannot handle this scenario in an optimized manner. The use of smart signal in this case will make the situation better by automatically adjusting the signal timer to ensure a smooth and continuous flow of traffic. SCATS (Sydney Coordinated Adaptive Traffic System)[1] and SCOOT (Split Cycle Offset Optimization Technique)[2] are analogous ATSC systems for urban traffic coordination and management. However cost of implementing SCAT and SCOOT per intersection is very high. Hence, we are proposing a smart signal system which uses reinforcement learning along with object detection for providing efficient traffic signal timings in response to real-time traffic scenario. In this paper we are proposing a complete working system which can be implemented at traffic intersections as a replacement for conventional systems. Our proposed system is substantially cheaper than the current systems in use, has fewer hardware and computational requirements.

2. Literature Review

A lot of research has been done on applications of reinforcement learning for adaptive traffic signal control.
We have referred the following research for our proposed system.
Thrope applied SARSA to a simulated traffic control problem. The use of this algorithm lowered the average vehicle timing by 29%.[3]
Abdulhai et Al applied the Q-Learning to a simple two-phased isolated traffic network. The isolation outperformed the pre-timed control scheme for variable traffic scenario by around 44%.[4]

A . Reinforcement Learning-

Reinforcement learning is an area of machine learning. It focuses on maximizing rewards in a given environment. Reinforcement learning differs from supervised learning in the sense that in supervised learning the algorithm is provided with test inputs and their corresponding outputs. On the contrary in reinforcement learning the agent learns through exploration and exploitation in the given environment. This is similar to how living beings adapt to their surrounding environments, hence making Reinforcement learning suitable for developing intelligent agents. Reinforcement learning comprises of at least one agent and an environment. The agent senses the environment's state and selects an action based on the current environment state and it's Knowledge. The environment then returns a reward, which may be positive or negative, to the agent. The agent then updates its Knowledge based on the received reward. Reinforcement learning has proved to provide near optimal solutions in stochastic environments like in the traffic signal problem. Hence, we are using reinforcement learning to compute optimal traffic signal timing policy in our proposed system.

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Q-Learning:
Q-Learning[5] is a variation of reinforcement learning. In Q-Learning we have a Q Function which takes input a state-action pair and returns long term expected reward value for taking the given action in that state. The formula for Q-Learning is given as follows-

$$Q^{new}(s_t, a_t) \leftarrow Q^{old}(s_t, a_t) + \alpha \cdot \left(r_t + \gamma \cdot maxQ(s_{t+1}, a) - Q^{old}(s_t, a_t)\right)$$

where,
α- Learning Rate
γ- Discount factor

Hence, while selecting the optimal action the agent can use greedy approach and select the action with maximum Q value for given state. In our proposed system Q-learning is used to compute the long term reduction in waiting time for vehicles at a traffic signal.

B . Object Detection

For the proposed system we used YOLO[6] (You Only Look Once) Real Time Object Detection. Earlier Image detection techniques such as R-CNN and Fast R-CNN use classifiers or localizers differently for detection. YOLO has less shortcomings as compared to the prior techniques. YOLO v3[7] is much faster, provides much more accurate results and can be trained for more than 200 classes. Moreover, it allows to make an easy tradeoff between speed and accuracy.

We used the COCO (Common Objects in Context) dataset to train a custom model for detecting the vehicles. The model is specifically designed to detect 4 classes namely (“cars”, “buses”, “trucks”, “people”). We use this object detection model on the image of the traffic in front of the signal so as to get the count of vehicles, which gives us the idea about the current traffic condition at that signal.

3. Proposed Work

Our proposed system is a complete working model comprising of the Reinforcement Learning and Object detection algorithms, hardware and other components. Functioning of each component is explained as follows:

- Actions -

The actions are taken by the agent according to the state of the Environment perceived by the agent. The actions in our RL agent are the possible green phase timings of the traffic signal. In our implementation, we defined the actions to be multiples of 10 from 10 to 100 included. These represent the time in seconds of the green phase.

- State Space Representation -

The State Space is the Mapping of State-Action pairs with their corresponding Q-values. In our implementation, the State space is represented by a Matrix. The dimensions of the Matrix are 100(Possible values of Traffic Density) x 10 (Number of possible actions). Each cell represents a Q-value for a pair of State and action that can be taken. The following figure represents the State Space in our system.

3.1 Reinforcement Learning
In our proposed system reinforcement learning is responsible for providing the green phase timing according to the current state of traffic and learn from the actions taken. The main components of the Reinforcement Learning algorithm are:

- Environment
- State
- Action
- Reward

Their implementation in the proposed system is explained in detail as follows -

- Environment -

The environment, in this case, is the Traffic in front of the Traffic signal. The RL agent percepts the environment using the image of the traffic in front of the signal captured by the camera. The agent interacts with the environment in the form of the green light timings.

- State -

The object detection algorithm returns the count of vehicles in the image captured by the camera in JSON format. The JSON data contains each class (E.g. “Cars”, “Bus”, “Truck”, etc) of the objects detected as the keys and their respective counts as the values to the keys. The state of the environment is represented by a variable called ‘Traffic Density’ which is calculated as a weighted sum of all counts of various types of vehicles by the object detection algorithm. The weights for the weighted average can be adjusted through further trails. In our system, we have assigned higher weights to vehicles who are large in size or have slower moving speeds, i.e. weights should be assigned to the specific class of vehicles according to their effect on the traffic.

In our implementation, we defined the states, i.e. possible values of traffic density as from 1 to 100.

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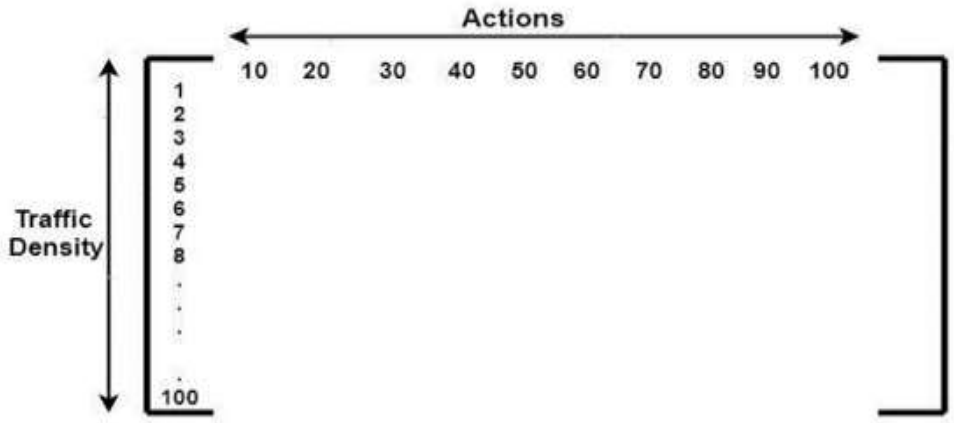

Fig.1-State Space Representation

- Rewards -
  Rewards are returned by the environment based on the effect of the actions i.e. Green phase timings taken by the RL agent.

In our implementation, the rewards are given in the following manner:

- ❑ Halfway through the green phase of the signal, the system again checks the state of the environment, if the traffic density is reduced to less than one-third of the previous state of the environment, the agent receives a positive reward of +1.
- ❑ Else if at the end of the green phase, the traffic density is less than its value in the previous state, the Agent receives a positive reward of +1.
- ❑ If both above cases are not satisfied, then the Agent receives a negative reward of -1.

3.2 Server

In our proposed system, the server plays an important part. As the raspberry pi (Microcontroller) has very limited computation power, it is not possible to run the object detection algorithm on it locally. Hence we hosted the object detection model on a cloud server[8-12]. We created an API to access the object detection model. The microcontroller sends an image to the model in the server -via the API and the server responds with the JSON data of object counts.

Deploying the object detection model on the cloud server has following advantages -

- It allows us to leverage the computational power of the cloud.
- It saves money, as multiple signals can use a single server via the API.

The computational power of the cloud server can be scaled as per the requirement.

3.3 Signal

The final component of the proposed system is the Traffic Signal itself. The system needs to be integrated with the existing traffic signal controllers, so that the green phase timings given by the RL agent, can be used to control the traffic lights.

3.4 Microcontroller

In our proposed system, the microcontroller is a central component of the system. The RL agent runs on the Microcontroller. The Microcontroller is also responsible for capturing the images of the traffic in front of the signal through the camera attached to it. It then sends the captured image to the cloud server via the API, once the server responds, it then sends the object detection data to the agent. Once the agent computes the green phase timing, the microcontroller sends this timing to the traffic signal.

We have used raspberry pi 3B+ microcontroller[9] because it provides sufficient computing capability and functionality for our system. The raspberry pi camera v2 is an ideal camera that is compatible with our setup and provides image quality that is suitable for the application.

The following figure shows detailed flow of working of the proposed system–

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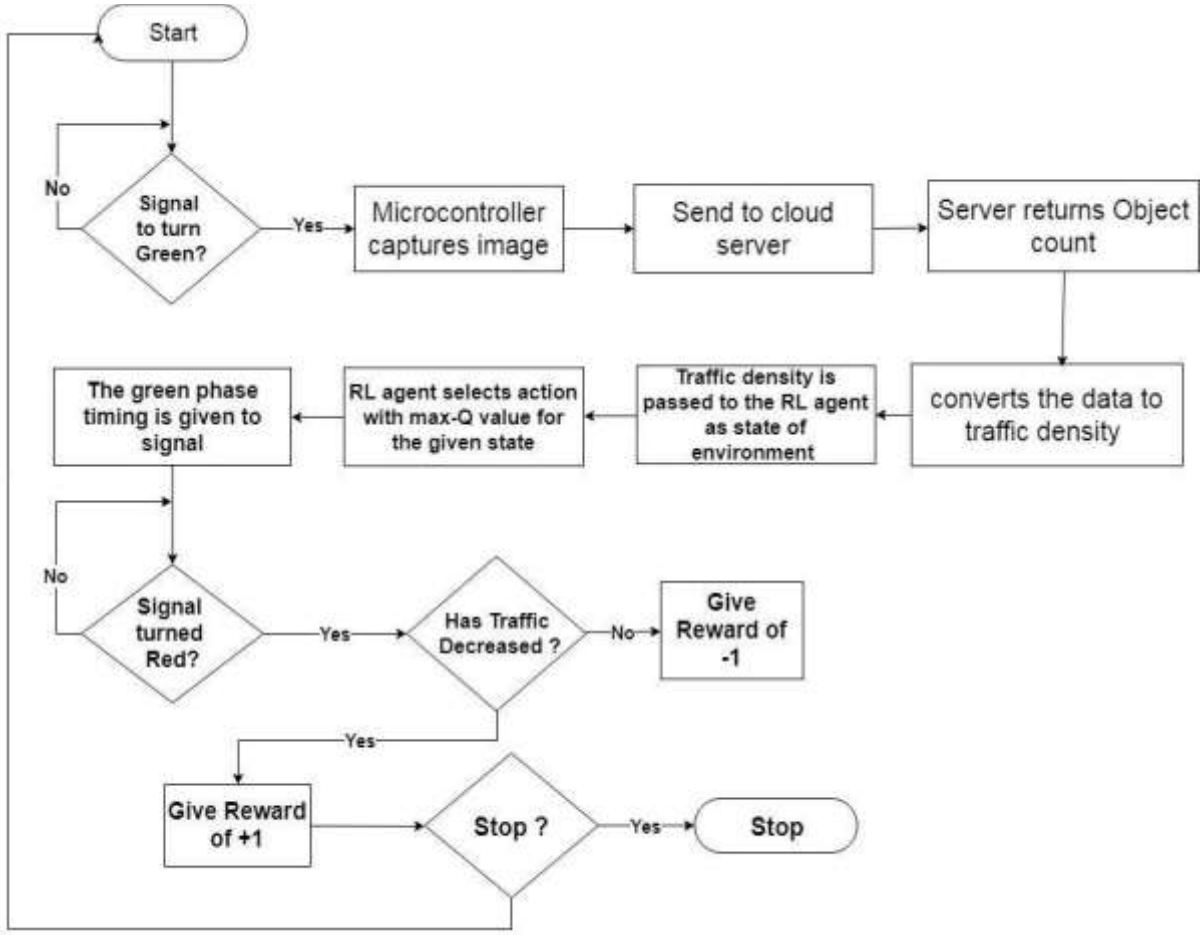

Fig.2-Proposed System flow chart

4. Results and Analysis

For testing the RL component, simulation was used. We used a Python script known as traffic_sim.py. It is written by Dr. Phillip M. Feldman[10]. It uses SimPy library in python. An environment class was created to act as an interface between the Simulation and the RL agent.

Results are reported for:

1. Simulation with no Reinforcement Algorithm.
2. Simulation with SARSA(State-Action-Reward-State-Action)
3. Simulation with Q-Learning.

	Mean No. of Cars Waiting	**Mean No. of Cars Waiting**
No Algorithm	6.621	32.803
SARSA	5.412	26.987
Q-Learning	4.079	20.476

Table.1- Results of Simulations

SARSA **:** In this approach we consider Epsilon(**ε**) equal to 0.7 i.e the exploration and exploitation constant. Using this approach we observe 23.58% decrease in mean number of cars waiting at any point of time and 24.55% decrease in Mean waiting time for cars.

Q-Learning: We ran our simulation for 5 simulation days for analysis of all possible combinations of Q-Learning parameters (Epsilon, Alpha and Gamma). After the simulation the best possible combination for the values of Epsilon (the exploration and exploitation constant), alpha

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(learning rate), gamma (discount factor) were equal to 0.7, 0.1 and 0.1 respectively.

Using this approach we observed 38.4% decrease in Mean number of cars waiting at any point of time and 37.6% decrease in Mean waiting time for cars.

In our proposed system the cost of the initial hardware installation for every signal is less than $100. In addition, a cloud server with decent configuration used for hosting object detection model can cost around $20 per month and can handle request from about 20 signals concurrently. Hence our system is a substantially economical than the current ATSC systems in use.

5. Conclusion and Future work

In this paper, proposed model completes a working system for adaptive traffic signal control. The proposed systemuses object detection for detecting real-time traffic scenario and reinforcement learning for deciding the proper green phase timing for that scenario. The proposed system was tested on simulation for a single traffic signal and it showed a substantial decrease in average waiting time for the vehicles at the signal. In addition, the cost of implementation of the proposed system is substantially lesser than the current systems in use, hence proving suitable for implementation in developing economies.

The proposed system can show better performance if multi-agent reinforcement learning is applied, whereby multiple agents can compute a shared policy by combining their acquired knowledge, hence providing more efficient traffic signal timings. Centralized control can be integrated with the proposed system, which can provide better control and monitoring to the traffic police or concerned authorities.

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